

Signal Conditioning with RC Filters

Measuring the gain and phase response of first-order low-pass and high-pass filters

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Course	ME 343-004 Mechanical Lab I, NJIT	Instructor	Prof. B. S. Mani · TA: Saba Mirahsani
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PROJECT SUMMARY

This experiment compared the measured and theoretical frequency response of first-order RC low-pass and high-pass filters. A function generator swept the input from 200 Hz to 40 kHz, a frequency counter verified each setting, and an oscilloscope recorded input and output amplitude and the phase shift between them. With $R = 1\text{ k}\Omega$ and $C = 80\text{ nF}$, the theoretical cutoff frequency is 1.99 kHz. Both filters behaved as predicted: the low-pass filter passed low frequencies and attenuated high ones, crossing -45° of phase right at the cutoff, while the high-pass filter did the reverse, with $+44^\circ$ of phase lead at 2 kHz.

1. Introduction & Theory

Filters shape a signal according to its frequency content, which makes them central to signal conditioning in any measurement system. A **low-pass** RC filter passes low frequencies while attenuating high ones, with the phase lag growing as frequency rises. A **high-pass** filter does the opposite: it blocks low-frequency content and passes high frequencies, with a phase lead that shrinks as frequency rises. Both are characterized by the cutoff frequency, where output power is halved (gain = 0.707, or -3 dB).

Table 1. First-order RC filter relationships

Quantity	Low-pass	High-pass
Cutoff frequency	$f_c = 1 / (2\pi RC)$	$f_c = 1 / (2\pi RC)$
Gain $ V_o/V_i $	$1 / \sqrt{1 + (f/f_c)^2}$	$(f/f_c) / \sqrt{1 + (f/f_c)^2}$
Phase shift	$\phi = -\tan^{-1}(f/f_c)$	$\phi = \tan^{-1}(f_c/f)$
Gain in decibels	$20 \log_{10}(V_o/V_i)$	$20 \log_{10}(V_o/V_i)$

2. Procedure

The function generator supplied a sinusoidal input to the RC filter, and the frequency counter confirmed the actual input frequency. Both the input and output were displayed on the oscilloscope, where the peak-to-peak amplitudes and the phase shift between them were read at eight frequencies from 200 Hz to 40 kHz. The procedure was repeated for the high-pass configuration.



Figure 1. Function generator

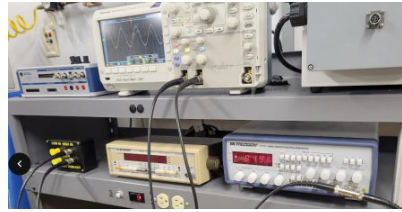


Figure 2. Experimental setup

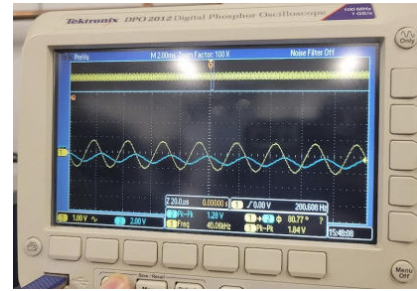


Figure 3. Oscilloscope capture of the low-pass response

3. Results

Table 2. Low-pass filter response

Frequency (Hz)	V_{in} p-p (V)	V_{out} p-p (V)	Gain (measured)	Gain (theory)	Phase (measured)	Phase (theory)
199.2	3.84	3.80	0.990	0.995	-7.4°	-5.7°
500.3	3.92	3.72	0.949	0.970	-14.7°	-14.1°
1,000	3.84	3.46	0.901	0.893	-23.5°	-26.7°
1,999	3.84	2.82	0.734	0.705	-42.0°	-45.1°
4,001	3.76	1.86	0.495	0.445	-57.5°	-63.6°
8,000	3.68	1.02	0.277	0.241	-70.7°	-76.0°
20,009	3.68	0.43	0.117	0.099	-83.1°	-84.3°
40,029	3.68	0.22	0.060	0.050	-85.4°	-87.2°

Table 3. High-pass filter response

Frequency (Hz)	V_{in} p-p (V)	V_{out} p-p (V)	Gain (measured)	Gain (theory)	Phase (measured)	Phase (theory)
200	3.92	0.41	0.105	0.100	79.7°	84.3°
500.9	3.92	0.84	0.214	0.244	74.6°	75.9°
1,005.5	3.92	1.51	0.385	0.451	63.5°	63.2°
2,004.8	3.84	2.04	0.531	0.710	44.0°	44.8°
3,999.4	3.84	3.18	0.828	0.895	31.1°	26.4°
8,004.6	3.76	3.52	0.936	0.970	17.0°	14.0°
20,002.9	3.68	3.60	0.978	0.995	8.7°	5.7°
40,004.3	3.68	3.64	0.989	0.999	2.3°	2.8°

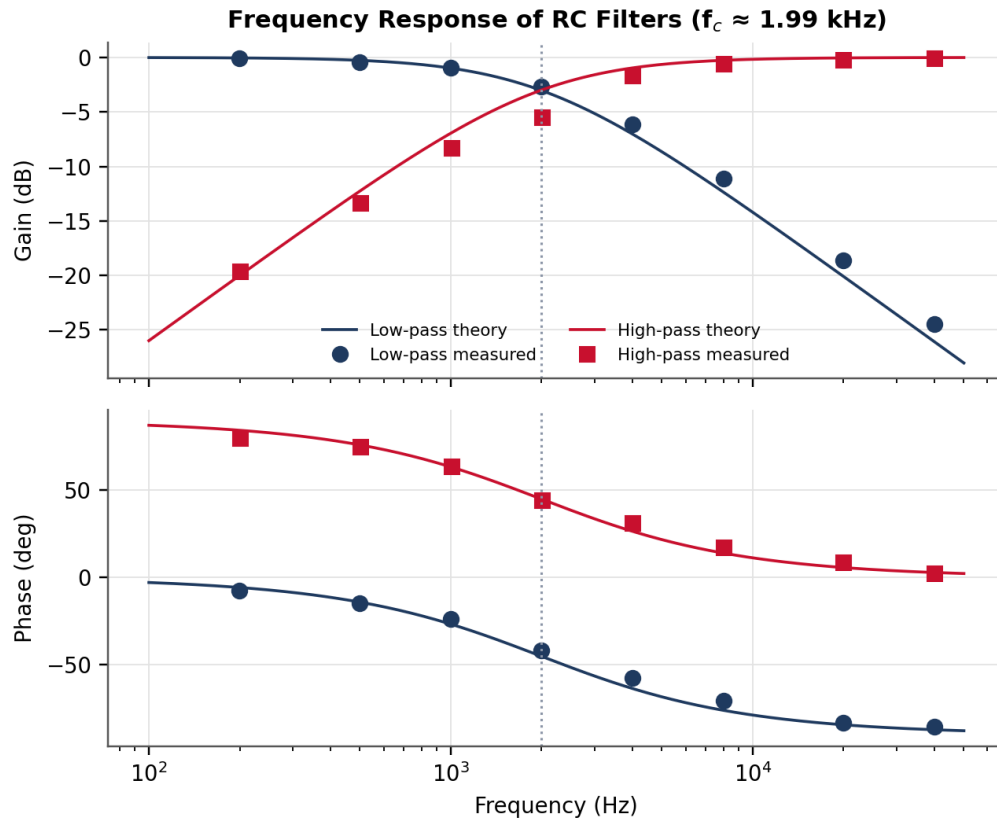


Figure 4. Bode plot of measured (markers) and theoretical (lines) gain and phase. The dotted line marks the 1.99 kHz cutoff.

4. Discussion

Both filters followed first-order RC theory closely across two decades of frequency. The **phase** results were especially strong: the low-pass filter measured -42° at 2 kHz and the high-pass filter $+44^\circ$, both close to the theoretical $\pm 45^\circ$ at the cutoff, which confirms the 1.99 kHz cutoff frequency set by R and C.

The **low-pass gain** matched theory within a few hundredths at nearly every point, rolling off toward the expected -20 dB per decade above the cutoff. The **high-pass gain** tracked the correct trend but ran below theory in the middle of the sweep, most noticeably at 2 kHz (0.53 measured vs. 0.71 predicted). Likely causes include component tolerances in R and C, the oscilloscope probe loading the circuit, and the difficulty of reading small peak-to-peak amplitudes precisely on the screen. The input amplitude also drifted slightly (3.92 to 3.68 V) across the sweep, which is why gain was calculated from the measured input at each frequency rather than a single nominal value.

5. Conclusion

The experiment confirmed that simple resistor–capacitor circuits can selectively shape signals by frequency. Measured cutoff behavior, gain roll-off, and phase shift for both the low-pass and high-pass filters closely matched theoretical predictions, reinforcing the role of frequency-dependent components in signal processing and conditioning systems.

6. Extension: Band-Pass Filter

A band-pass filter can be built from a series R_1 – C_1 branch (which blocks low frequencies) feeding a parallel R_2 – C_2 output (which shunts high frequencies). Using complex impedances $Z_1 = R_1 + 1/(j\omega C_1)$ and $Z_2 = R_2 / (1 + j\omega R_2 C_2)$, the voltage divider $V_o/V_i = Z_2/(Z_1 + Z_2)$ gives a gain magnitude of:

$$|V_o/V_i| = 1 / \sqrt{[1 + C_2/C_1 + R_1/R_2]^2 + (2\pi f C_2 R_1 - 1/(2\pi f C_1 R_2))^2}$$

with phase shift $\phi = -\tan^{-1} [(\omega C_2 R_1 - 1/(\omega C_1 R_2)) / (1 + C_2/C_1 + R_1/R_2)]$. The gain peaks where the two frequency terms cancel and falls off on either side.

References

- Hatim, N. (2017). *Signal conditioning by RC filter* [Laboratory manual]. ME 343 Laboratory, New Jersey Institute of Technology.
- Zhu, C. (2009). *ME 343 laboratory introduction*. New Jersey Institute of Technology.